

Apache Virtual Host Configuration (IP-Based Hosting)

1. **Check the Change Request:** Verify if the change is assigned and check its status (Approved/Pending). Start work only after approval.
2. **login :** Get the server IP from the TL and log in using SSH, RDP, VNC, or Remote Console.
- 3.**Take Backup:** Always take a backup before changes. Use NetBackup for physical servers or snapshot for virtual machines.

4.Configure Network and Hostname

- Set network adapter to bridged (automatic) and configure static IP using:

```
# nmcli con mod ens33 ipv4.addresses "192.168.0.100/24" ipv4.gateway "192.168.0.1"
```

```
# hostnamectl set-hostname radical.com
```

```
[root@dnssrv httpd]# hostnamectl set-hostname www.radical.com
[root@dnssrv httpd]#
[root@dnssrv httpd]#
[root@dnssrv httpd]#
[root@dnssrv httpd]# hostname
www.radical.com
[root@dnssrv httpd]#
f[root@dnssrv httpd]#
[root@dnssrv httpd]# ifconfig
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.0.100 netmask 255.255.255.0 broadcast 192.168.0.255
    inet6 2401:4900:8925:e6e3:20c:29ff:fe6b:c58a prefixlen 64 scopeid 0x0<global>
    inet6 fe80::20c:29ff:fe6b:c58a prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:6b:c5:8a txqueuelen 1000 (Ethernet)
    RX packets 15763 bytes 1605356 (1.5 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 11594 bytes 1176157 (1.1 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

5.Configure Repository (Verify and Install Required Packages)

```
#yum search httpd
#yum info httpd
#yum install httpd.x86_64
```

```
[root@dnssrv /]# yum search httpd
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use subscription-manage
this repo is file based for insatlling packages
this repo is file based for insatlling packages
Last metadata expiration check: 0:00:01 ago on Wed 27 May 2026 01:09:45 PM IST.
===== Name Exactly Matched: httpd =====
httpd.x86_64 : Apache HTTP Server
===== Name & Summary Matched: httpd =====
```

```
[root@dnssrv ~]# yum install httpd.x86_64
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use subscription-manager to register.

Last metadata expiration check: 0:00:54 ago on Wed 27 May 2026 01:09:45 PM IST.
Dependencies resolved.
=====
Package                                Architecture      Version           Repository
=====
Installing:
httpd                                   x86_64            2.4.53-11.el9_2.4 AppStream
Installing dependencies:
```

6. After installing httpd, next step is to go to /var/www/html/ to create your web page, because this is the default document root of Apache.

Simple explanation:/var/www/html/ is the location where your website files are stored

```
#cd /var/www/html/
#vi index.html
#cat index.html
#httpd -t=syntax for error checking
```

```
[root@www html]#
[root@www html]# cd /var/www/html
[root@www html]#
[root@www html]#
[root@www html]#
[root@www html]# vi index.html
[root@www html]#
[root@www html]#
[root@www html]# cat index.html
***** first sample website *****
[root@www html]#
[root@www html]# httpd -t
Syntax OK
```

7. Create Virtual Host file

Location of Virtual Host File:

/etc/httpd/conf.d/

Why this location?

Apache automatically reads all .conf files from /etc/httpd/conf.d/, so any virtual host configuration placed here gets applied without modifying the main httpd.conf.

Purpose of Virtual Host: We create a Virtual Host file to host multiple websites on a single server.

Concept of virtual host:

- One Server (One Hardware + One OS)
- Multiple IPs / Multiple Domains
- Multiple Websites running simultaneously

8. Find that configuration file.

```
[root@dnssrv /]# rpm -qc httpd.x86_64
/etc/httpd/conf.modules.d/00-brotli.conf
/etc/httpd/conf.modules.d/00-systemd.conf
[root@dnssrv /]#
[root@dnssrv /]#
[root@dnssrv /]#
[root@dnssrv /]#
[root@dnssrv /]# vi /etc/httpd/conf/httpd.conf
[root@dnssrv /]#
```

rpm -qc httpd.x86_64 shows only the configuration files that are marked by the package. That is why you see files from /etc/httpd/conf.modules.d/.

The file /etc/httpd/conf/httpd.conf is the main configuration file, but sometimes it may not appear in rpm -qc output depending on how the package defines its config files

9. Take backup in tmp location:

```
cp /etc/httpd/conf/httpd.conf /tmp/httpd.conf.bkp1
```

10..Go to location **cd /etc/httpd/conf.d/** create **multiplewebsite.conf** file to define Virtual Host configuration for hosting multiple websites

```
[root@www conf.d]# cd /etc/httpd/conf.d/
[root@www conf.d]#
[root@www conf.d]#
[root@www conf.d]#
[root@www conf.d]# vi multiplewebsite.conf
[root@www conf.d]#
[root@www conf.d]#
```

Make these changes(To protect page using Virtual Host + username/password)

Create password file `htpasswd -c /etc/httpd/.htpasswd user1`

```
[root@www conf.d]#
[root@www conf.d]#
[root@www conf.d]# htpasswd -c /etc/httpd/.htpasswd aarti
New password:
Re-type new password:
```

cd /etc/httpd/conf.d/multiplewebsite.conf

```
root@dnssrv:/etc/httpd/conf.d — /usr/bin/vim multiplewebsite.conf

root@dnssrv:/var/www/html x root@dnssrv:/etc/httpd/conf.d — /usr/bin/vim multiplewebsite.conf x

<VirtualHost 192.168.0.100:80>
    ServerAdmin root@radical.com
    ServerName radical.com
    ServerAlias www.radical.com
    DocumentRoot /var/www/html
    ErrorLog /var/log/httpd/www.radical.com.error.log
    CustomLog /var/log/httpd/www.radical.com.access.log combined
    <Directory "/var/www/html">
        AuthType Basic
        AuthName "Restricted Area"
        AuthUserFile /etc/httpd/.htpasswd
        Require valid-user
    </Directory>
</VirtualHost>
```

Start stop service and check logs for better understanding

```
[root@www conf.d]#
[root@www conf.d]# systemctl start httpd
[root@www conf.d]#
[root@www conf.d]#
[root@www conf.d]#
[root@www conf.d]# systemctl stop httpd
[root@www conf.d]#
[root@www conf.d]#
[root@www conf.d]#
[root@www conf.d]# systemctl enable --now httpd
[root@www conf.d]#
[root@www conf.d]#
[root@www conf.d]#
[root@www conf.d]#
[root@www conf.d]# systemctl restart httpd
[root@www conf.d]#
[root@www conf.d]#
```

```
[root@www html]#
[root@www html]# tail -f /var/log/messages
May 27 16:50:01 dnssrv NetworkManager[1066]: <info> [177980001.7112] agent-manager: agent[02cb743e20bac8b0,:1.67/org.gnome.Shell.Ne
tworkAgent/0]: agent registered
May 27 16:50:27 dnssrv systemd[1]: fprintd.service: Deactivated successfully.
May 27 17:03:33 dnssrv cupsd[1076]: REQUEST localhost - - "POST / HTTP/1.1" 200 183 Renew-Subscription client-error-not-found
May 27 17:17:02 dnssrv systemd[1]: Starting dnf makecache...
May 27 17:17:03 dnssrv dnf[11592]: Updating Subscription Management repositories.
May 27 17:17:03 dnssrv dnf[11592]: Unable to read consumer identity
May 27 17:17:03 dnssrv dnf[11592]: This system is not registered with an entitlement server. You can use subscription-manager to regi
ster.
May 27 17:17:03 dnssrv dnf[11592]: Metadata cache refreshed recently.
May 27 17:17:03 dnssrv systemd[1]: dnf-makecache.service: Deactivated successfully.
May 27 17:17:03 dnssrv systemd[1]: Finished dnf makecache.
=====May 27 17:17:44 dns
srv gnome-shell[2227]: Window manager warning: last_user_time (18386757) is greater than comparison timestamp (18386726). This most
likely represents a buggy client sending inaccurate timestamps in messages such as _NET_ACTIVE_WINDOW. Trying to work around...
May 27 17:17:44 dnssrv gnome-shell[2227]: Window manager warning: W31 appears to be one of the offending windows with a timestamp of
18386757. Working around...
=====
May 27 17:17:55 dnssrv systemd[1]: Stopping The Apache HTTP Server...
May 27 17:17:56 dnssrv systemd[1]: httpd.service: Deactivated successfully.
May 27 17:17:56 dnssrv systemd[1]: Stopped The Apache HTTP Server.
May 27 17:17:56 dnssrv systemd[1]: httpd.service: Consumed 6.556s CPU time.
May 27 17:17:59 dnssrv systemd[1]: Starting The Apache HTTP Server...
May 27 17:17:59 dnssrv httpd[11663]: Server configured, listening on: port 80
May 27 17:17:59 dnssrv systemd[1]: Started The Apache HTTP Server.
May 27 17:18:11 dnssrv gnome-shell[2227]: Window manager warning: last_user_time (18413534) is greater than comparison timestamp (184
```

11.DNS configuration for name resolution.

```
root@dnssrv:/etc/httpd/conf.d — /usr/bin/vim /var/named/fzone.radical.com
root@dnssrv:/var/www/html — tail -f /var/log/messages
root@dnssrv:/etc/httpd/conf.d — /usr/bin/vim /var/named/fzone.ra...

$TTL 1D
@      IN SOA  dnssrv.radical.com. root.dnssrv.radical.com. (
                                0      ; serial
                                1D     ; refresh
                                1H     ; retry
                                1W     ; expire
                                3H )   ; minimum

@      NS   dnssrv.radical.com.
www    A    192.168.0.100
dnssrv A    192.168.0.100
```

```
[root@www conf.d]# named-checkzone radical.com /var/named/fzone.radical.com
zone radical.com/IN: loaded serial 0
OK
[root@www conf.d]#
[root@www conf.d]#
[root@www conf.d]#
[root@www conf.d]# systemctl start named
[root@www conf.d]#
[root@www conf.d]#
[root@www conf.d]#
[root@www conf.d]# systemctl restart named
```

12.Temporarily disable Firewall for remote checking

```
[root@www conf.d]# getenforce
Permissive
[root@www conf.d]#
[root@www conf.d]#
[root@www conf.d]# setenforce 0
[root@www conf.d]#
[root@www conf.d]#
[root@www conf.d]#
[root@www conf.d]# systemctl stop firewalld.service
```

13.Test from remote machine

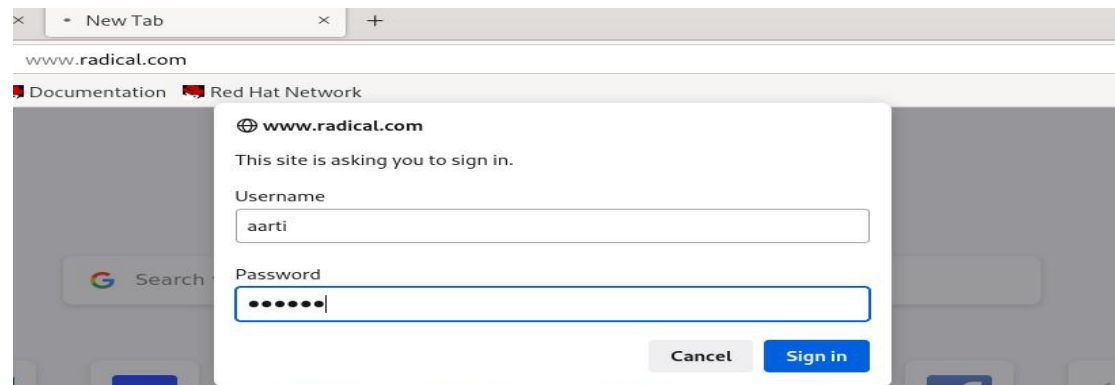
using client IP 192.168.0.101 on bridged network.

DNS server IP is set in /etc/resolved.conf for name resolution.

```
[root@dhcpsrv1 ~]# vi /etc/resolv.conf
[root@dhcpsrv1 ~]#
[root@dhcpsrv1 ~]#
[root@dhcpsrv1 ~]#
[root@dhcpsrv1 ~]# cat /etc/resolv.conf
# Generated by NetworkManager
search radical.com
nameserver 192.168.0.100
[root@dhcpsrv1 ~]#
[root@dhcpsrv1 ~]#
```


Enter the website domain name in the browser 

Asking for password



The website is displayed using its domain name



14.since it is working, allow the firewall rule permanently and test again.

```
[root@www conf.d]#  
[root@www conf.d]# systemctl start firewalld.service  
[root@www conf.d]#  
[root@www conf.d]# firewall-cmd --add-service=http  
success  
[root@www conf.d]# firewall-cmd --reload  
success  
[root@www conf.d]#  
[root@www conf.d]# netstat -antlpu | grep httpd  
tcp6      0      0 :::80          :::*           LISTEN    13440/httpd  
[root@www conf.d]#  
[root@www conf.d]#  
[root@www conf.d]#  
[root@www conf.d]#
```

14.Now you can add multiple IPs and multiple webpages using Virtual Host configuration 

```
[root@www html]# nmcli con modify ens33 +ipv4.addresses "192.168.0.102/24"  
[root@www html]#  
[root@www html]# nmcli con modify ens33 +ipv4.addresses "192.168.0.103/24"  
[root@www html]# nmcli con up ens33  
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/11)  
[root@www html]# ip a s  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host  
        valid_lft forever preferred_lft forever  
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 00:0c:29:6b:c5:8a brd ff:ff:ff:ff:ff:ff  
    altname enp2s1  
    inet 192.168.0.100/24 brd 192.168.0.255 scope global noprefixroute ens33  
        valid_lft forever preferred_lft forever  
    inet 192.168.0.102/24 brd 192.168.0.255 scope global secondary noprefixroute ens33  
        valid_lft forever preferred_lft forever  
    inet 192.168.0.103/24 brd 192.168.0.255 scope global secondary noprefixroute ens33  
        valid_lft forever preferred_lft forever  
    inet6 fe80::20c:29ff:fe6b:c58a/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever
```

Cd/etc/httpd/conf.d/ 

```
<VirtualHost 192.168.0.100:80>
ServerAdmin root@radical.com
ServerName radical.com
ServerAlias www.radical.com
DocumentRoot /var/www/html
ErrorLog /var/log/httpd/www.radical.com.error.log
CustomLog /var/log/httpd/www.radical.com.access.log combined
<Directory "/var/www/html">
    AuthType Basic
    AuthName "Restricted Area"
    AuthUserFile /etc/httpd/.htpasswd
    Require valid-user
</Directory>
</VirtualHost>

<VirtualHost 192.168.0.102:80>
ServerAdmin root@yahoo.com
ServerName yahoo.com
ServerAlias www.yahoo.com
DocumentRoot /var/www/html/yahoo
ErrorLog /var/log/httpd/www.yahoo.com.error.log
CustomLog /var/log/httpd/www.yahoo.com.access.log combined
</VirtualHost>

<VirtualHost 192.168.0.103:80>
ServerAdmin root@aarti.com
ServerName aarti.com
ServerAlias www.aarti.com
DocumentRoot /var/www/html/aarti
ErrorLog /var/log/httpd/www.aarti.com.error.log
CustomLog /var/log/httpd/www.aarti.com.access.log combined
</VirtualHost>
```

Then add DNS records in forward zone 

yahoo A 192.168.0.102

aarti A 192.168.0.103

```
root@dnssrv:/var/www/html

$TTL 1D
@      IN SOA  dnssrv.radical.com. root.dnssrv.radical.com. (
                                0      ; serial
                                1D     ; refresh
                                1H     ; retry
                                1W     ; expire
                                3H )   ; minimum

@      NS   dnssrv.radical.com.
www    A    192.168.0.100
dnssrv A    192.168.0.100
www.yahoo.com A 192.168.0.102
yahoo  A    192.168.0.102
aarti  A    192.168.0.103
www.aarti.com A 192.168.0.103
```

Check again from the client side 

```
[root@dhcpsrv1 ~]# curl yahoo
***** this is www.yahoo.com *****
[root@dhcpsrv1 ~]#
[root@dhcpsrv1 ~]#
[root@dhcpsrv1 ~]# curl www.yahoo.com
***** this is www.yahoo.com *****
[root@dhcpsrv1 ~]#
[root@dhcpsrv1 ~]# curl aarti
***** this is my website newly created *****
```

Name-Based Virtual Hosting Configuration

Name-based hosting is used to host multiple websites on a single server using one IP address.

1. remove previously added IP aliasing/static IPs from ens33, use:

```
#nmcli con modify ens33 -ipv4.addresses "192.168.0.102/24"
```

```
#nmcli con modify ens33 -ipv4.addresses "192.168.0.103/24"
```

```
syntax OK
[root@www html]# nmcli con modify ens33 -ipv4.addresses "192.168.0.102/24"
[root@www html]#
[root@www html]#
[root@www html]# nmcli con modify ens33 -ipv4.addresses "192.168.0.103/24"
[root@www html]#
[root@www html]#
[root@www html]#
[root@www html]#
```

2. Remove IPs from all <VirtualHost> entries and use only:<VirtualHost *:80>
Your corrected multiplewebsite.conf should look like this: 

```
root@dnssrv:/etc/httpd/conf.d — /usr/bin/vim multiplewebsite.conf
root@dnssrv:/var/www/html
root@dnssrv:/etc/httpd/conf.d — /usr/bin/vim multiplewebsite.conf
<VirtualHost *:80>
    ServerAdmin root@radical.com
    ServerName radical.com
    ServerAlias www.radical.com
    DocumentRoot /var/www/html
    ErrorLog /var/log/httpd/www.radical.com.error.log
    CustomLog /var/log/httpd/www.radical.com.access.log combined
    <Directory "/var/www/html">
        AuthType Basic
        AuthName "Restricted Area"
        AuthUserFile /etc/httpd/.htpasswd
        Require valid-user
    </Directory>
</VirtualHost>
<VirtualHost *:80>
    ServerAdmin root@yahoo.com
    ServerName yahoo.com
    ServerAlias www.yahoo.com
    DocumentRoot /var/www/html/yahoo
    ErrorLog /var/log/httpd/www.yahoo.com.error.log
    CustomLog /var/log/httpd/www.yahoo.com.access.log combined
</VirtualHost>
<VirtualHost *:80>
    ServerAdmin root@sartl.com
    ServerName sartl.com
    ServerAlias www.sartl.com
    DocumentRoot /var/www/html/sartl
    ErrorLog /var/log/httpd/www.sartl.com.error.log
```

- 3..Check syntax and restart service

```
[root@www conf.d]# httpd -t
Syntax OK
[root@www conf.d]#
[root@www conf.d]#
[root@www conf.d]# httpd -t
Syntax OK
[root@www conf.d]# systemctl restart httpd
[root@www conf.d]#
```


4.vi /var/named/fzone.radical.com

Add/edit like this for same IP name resolution:

```
root@dnssrv:/etc/httpd/conf.d — /usr/bin/vim /var/named/fzone.radical.com
root@dnssrv:/var/www/html
$TTL 1D
@ IN SOA dnssrv.radical.com. root.dnssrv.radical.com. (
    0 ; serial
    1D ; refresh
    1H ; retry
    1W ; expire
    3H ) ; minimum
@ NS dnssrv.radical.com.
www A 192.168.0.100
dnssrv A 192.168.0.100
www.yahoo.com A 192.168.0.100
www.aarti.com A 192.168.0.100
```

Check syntax: and restart

named-checkzone radical.com /var/named/fzone.radical.com

```
[root@www html]#
[root@www html]# named-checkzone radical.com /var/named/fzone.radical.com
zone radical.com/IN: loaded serial 0
OK
[root@www html]#
[root@www html]#
[root@www html]# systemctl restart named
[root@www html]#
```

5. Test from client:

First add dns ip in etc/resolv.conf

```
[root@dhcp1 ~]# vi /etc/resolv.conf
[root@dhcp1 ~]#
[root@dhcp1 ~]# cat /etc/resolv.conf
# Generated by NetworkManager
search radical.com
nameserver 192.168.0.100
```

6.curl -u aarti:redhat www.radical.com

curl www.yahoo.com

curl www.aarti.com

```
[root@dhcp1 ~]# curl www.aarti.com
***** this is my website newly created *****
[root@dhcp1 ~]#
[root@dhcp1 ~]#
[root@dhcp1 ~]# curl -u aarti:redhat www.radical.com
***** first sample website *****
[root@dhcp1 ~]#
[root@dhcp1 ~]#
[root@dhcp1 ~]#
[root@dhcp1 ~]# curl www.yahoo.com
***** this is www.yahoo.com *****
```

7. Open Firefox or any browser and test:

